

BST Working Paper — Volume 3: The Physics — Chapter 4: Cosmic Architecture

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Section 22: Antimatter, the Arrow of Time, and the Second Law

22.1 Commitment Order as Time

Time in BST is the direction of contact commitment on the substrate. Contacts commit — transitioning from superposition (uncommitted, quantum) to definite configuration (committed, classical) — and this commitment is irreversible without external work. The sequence of commitments defines a partial ordering on the contact graph. This ordering IS time. Not a parameter. Not a background. The physical process of the substrate becoming definite.

The second law of thermodynamics follows immediately. Entropy is $S = k \ln W$, where W counts the substrate configurations consistent with the current macrostate. Each commitment reduces the number of possible configurations (the committed contact is now definite) while increasing the number of committed contacts (the macrostate has more determined structure). The entropy of the macrostate increases because each commitment converts one degree of substrate freedom into one piece of macroscopic information. The conversion is one-way because commitment is one-way.

The arrow of time and the second law are the same principle: contacts commit and do not uncommit. There is no separate “past hypothesis” needed to explain why entropy was low at the Big Bang. The Big Bang was the phase transition from the pre-spatial state (fully connected, fully symmetric, maximum substrate entropy) to the spatial state (locally connected, symmetry broken, low macroscopic entropy). The macroscopic entropy was low because the phase transition had just begun — few contacts committed, little macroscopic structure, enormous remaining freedom. The subsequent increase of macroscopic entropy is the ongoing process of contact commitment — the universe becoming definite, one contact at a time.

22.2 The Absence of Time in BST’s Native Language

A structural feature of BST deserves explicit comment. The theory’s natural outputs are dimensionless ratios and energies — never durations:

- $m_p/m_e = 6\pi^5$ (dimensionless)
- $\sin^2 \theta_W = 3/13$ (dimensionless)
- $\alpha = 1/137.036$ (dimensionless)
- $T_{\text{deconf}} = \pi^5 m_e$ (energy)
- All Chern class coefficients $\{5, 11, 13, 9, 3\}$ (integers)

Every BST derivation that produces a time — the cosmic age $t_0 = 13.718$ Gyr, the neutron lifetime $\tau_n = 878.1$ s, the gravitational wave frequency $f = 6.4$ nHz, the Hubble constant H_0 — requires converting from energy units via $\hbar$ or c . Time is never the native output. It must be forced in.

This is not an oversight. It is the architecture. The substrate has no clock. It has topology (the contact graph), geometry (the bounded domain), and energy (the Bergman spectrum). Duration is what commitment looks like to an observer already embedded in the system — a derived quantity, not a fundamental one. The substrate knows

how many commitments have occurred and *what* their topology is. It does not know *how long* they took, because “how long” is a question asked in the language of time, and time is the answer, not the question.

This explains why BST’s most precise results are mass ratios and mixing angles (dimensionless, time-free), while quantities involving explicit time units tend to require additional physical input (the commitment rate, the expansion history). The theory speaks geometry. Time is a translation.

22.3 Antimatter as Anti-Commitment-Order Winding

A particle is a winding on S^1 aligned with the commitment direction — a circuit that propagates forward in the causal ordering of the contact graph. An antiparticle is a winding that opposes the commitment direction — a circuit propagating backward in the causal ordering.

This gives precise physical content to the Feynman-Stueckelberg interpretation, which treats antiparticles as particles moving backward in time. In standard QFT this is a mathematical convenience with no physical mechanism. In BST “backward in time” means “against the commitment order” — a winding on S^1 that opposes the direction in which contacts are committing. The interpretation becomes a mechanism.

22.4 CPT Invariance and CP Violation

CPT invariance follows from the structure of the contact graph. Reversing charge (flipping winding direction on S^1), parity (flipping spatial orientation on S^2), and time (flipping commitment order) together restores the original relationship between winding direction and causal direction. CPT invariance is the statement that physics depends on the relationship between these directions, not on their absolute orientations.

CP violation follows from the fact that the causal direction is physically real. Flipping the winding direction and the spatial orientation without flipping the commitment order changes the relationship between winding and causal direction. The resulting physics differs because the causal direction is a physical feature of the contact graph, not a convention.

The CKM phase — the single complex parameter responsible for all observed CP violation in the quark sector — arises from the complex structure of D_{IV}^5 . Real symmetric domains have no natural complex phases. Complex symmetric domains do. D_{IV}^5 is complex, so CP violation is built into the domain geometry. The magnitude of the CKM phase is determined by the specific complex structure of the domain — a geometric property, not a free parameter. The CKM mixing angles are now derived (Section 7.7): the Cabibbo angle $\sin \theta_C = 2/\sqrt{79} = 0.22502$ (0.004% from PDG, T1444 vacuum subtraction: $\text{rank}^4 n_C - 1 = 79$), the Wolfenstein parameter $A = (n_C - 1)/n_C = 4/5$, and $|V_{cb}| = A\lambda^2 = 4/125 = 0.0400$ (2.7% from PDG).

22.5 The Matter-Antimatter Asymmetry

During the pre-spatial phase transition, the symmetry between forward and backward causal directions was broken. The nucleation event defined a commitment direction. From that moment, forward windings (matter) and backward windings (antimatter) were no longer equivalent.

A forward winding propagates into uncommitted substrate — fresh contacts ready to commit. The winding proceeds with low impedance. A backward winding propagates against the commitment direction — attempting to wind through contacts that resist being organized opposite to their commitment. The backward winding encounters a slight impedance mismatch.

The impedance difference is tiny — almost negligible compared to the total winding energy. But it biases the production of forward windings over backward windings during the hot, dense conditions following the phase transition. When the universe cooled enough for matter-antimatter annihilation to complete, the slight excess of forward windings survived. This excess is the baryonic matter content of the universe.

The observed baryon-to-photon ratio $\eta \approx 6 \times 10^{-10}$ (approximately one excess baryon per billion baryon-antibaryon pairs) should be derivable from the critical exponents of the phase transition on D_{IV}^5 . The asymmetry near the critical point scales as a power of the order parameter (contact commitment density), with the power determined by the domain geometry.

Result (March 2026, updated April 2026): The baryon asymmetry is now derived: $\eta_b = (3/14)\alpha^4 = N_c/(2g) \times \alpha^4$ (T929), with four electromagnetic vertices and the color-over-genus prefactor $N_c/(2g) = 3/14$. This matches the Planck value $(6.104 \pm 0.058) \times 10^{-10}$ to 0.45%. The earlier route $\eta = 2\alpha^4/(3\pi)(1+2\alpha)$ (0.023%) required a radiative correction; the T929 form is cleaner — pure BST integers times α^4 . This removes η from the list of unexplained initial conditions. Full derivation: `notes/BST_BaryonAsymmetry_Eta.md`.

22.6 Why There Is Something Rather Than Nothing

The standard cosmological account has no principled explanation for why the universe contains matter. The Sakharov conditions (baryon number violation, CP violation, departure from equilibrium) identify necessary conditions for an asymmetry but do not determine its magnitude. Every baryogenesis mechanism in standard physics requires beyond-Standard-Model physics with tuned parameters.

BST's answer is structural. Matter exists because the universe has a time direction. The time direction biases forward windings. Forward windings are matter. The magnitude of the bias is determined by the geometry of the domain on which the phase transition occurs. No tuning. No new physics. Just the fact that a phase transition on a complex domain with a definite causal direction produces a slight preference for windings aligned with that direction.

One chain: D_{IV}^5 is complex $\rightarrow$ domain has natural complex phases $\rightarrow$ CP violation is geometric $\rightarrow$ phase transition establishes commitment direction $\rightarrow$ forward windings slightly favored $\rightarrow$ matter exceeds antimatter $\rightarrow$ we exist.

Section 23: The Wavefront and Computational Architecture

23.1 Changes, Not State

The commitment wavefront writes changes only. Each step, one contact transitions from uncommitted to committed. Its S^1 phase becomes definite. The previously committed contacts persist without re-execution. The 3D world at any moment is the integral of all prior commitments. The process at each step is the differential — one new phase appended to the accumulated structure.

The universe is an append-only log. Each commitment is a log entry. No random access. No rewrites. No deletes. Forward only. The arrow of time is not merely reflected in this architecture — it IS this architecture. The data structure permits only appending, so time can only advance.

23.2 Information per Commitment

One contact commitment determines one S^1 phase. The phase is constrained by neighboring committed contacts through holonomy requirements, Z_3 closure, and Haldane exclusion. The information content per commitment is the number of genuinely free bits after constraints are satisfied.

In sparse regions (low channel utilization): most of 137 slots are empty. Many phases are allowed. Information per commitment is roughly $\log_2(137)$ minus constraint reduction — approximately 5 to 7 genuinely free bits.

In dense regions (high channel utilization): exclusion constraints eliminate most options. Information per commitment drops toward zero. At channel saturation (black hole), each new commitment has essentially one allowed phase — the constraints fully determine the outcome. Zero free bits. This is why black holes are maximum entropy: each commitment carries no new information because the constraints leave no freedom.

In the pre-spatial state: zero commitments, zero macrostate information, maximum microstate freedom. The phase transition begins writing bits of the universe into existence, a few bits per commitment, from a blank substrate.

23.3 Projection: State from History

The 3D geometry at any moment is determined by the full set of committed contacts — all accumulated holonomies and committed phases. In this sense the projection reads the entire committed substrate to define the current metric.

But the dynamics at each moment depend only on the current committed state and its local neighborhood, not on how that state was reached. The projection for physics is local and present-tense. This is why physics is Markovian — the future depends on the present configuration of committed contacts, not on the history of commitments that produced it.

The substrate accumulates history (every commitment is permanent). The projection reads state (the current configuration). The dynamics depend on state, not history. The log is append-only. The query reads the materialized view.

23.4 Parallelism and Throughput

Two commitments can occur simultaneously if they are causally disconnected — if neither commitment's output affects the other's constraints. In sparse regions with low contact density, many commitments proceed in parallel. In dense regions, causal coupling forces sequential processing.

The universe's computational throughput — commitments per unit time — is determined by the parallelism available at the wavefront. This parallelism depends on the local constraint density, which depends on the matter-energy distribution.

Gravitational time dilation as write bottleneck: Dense contact regions near massive objects have more causal coupling between neighboring contacts. More coupling means less parallelism. Less parallelism means fewer simultaneous commitments per external time step. The local clock — the commitment rate — runs slower. Gravitational time dilation is not a geometric curiosity. It is a computational bottleneck caused by constraint density.

23.5 Total Information Budget

The observable universe has approximately 10^{122} Planck areas of horizon surface (Bekenstein bound). Each Planck area represents roughly one substrate contact. The total information content is $\sim 10^{122}$ bits. The wavefront writes at most 10^{122} bits per Planck time at maximum parallelism, reduced by causal coupling. The effective throughput is estimated at $\sim 10^{120}$ operations over the age of the universe, consistent with Lloyd's independent estimate of the universe's computational capacity.

23.6 Architecture Summary

The computational architecture of reality maps precisely onto a well-known engineering pattern:

Objects are particles — persistent topological configurations with defined properties (mass, charge, spin) and behaviors (interactions, decay channels). They maintain identity through topological error correction.

The append-only log is the commitment history — the sequence of contact commitments that cannot be reversed. Each entry records one phase determination. The log defines the arrow of time.

The materialized view is the 3D world — the current state reconstructed from the accumulated log. It is queried locally by the dynamics at each point but is not recomputed globally at each step.

The query engine is physics — the local rules (holonomy, exclusion, Z_3 closure) that determine the next log entry from the current local state. The rules are the same everywhere (the geometry of D_{IV}^5) but the results vary because the local state varies.

Causal consistency is maintained without a global coordinator. Each commitment references only its causal neighbors. Consistency follows from the topological constraints, not from a central authority.

Write throughput varies with constraint density. Sparse regions (voids) have high parallelism and fast clocks. Dense regions (near massive objects) have low parallelism and slow clocks. The throughput variation IS gravitational time dilation.

This is not an analogy. The universe is a distributed system maintaining consistent state through local append-only operations with causal consistency constraints. The architecture is the only one that works for this class of problem — no central coordinator, finite communication bandwidth, consistency required. Any engineer who has built distributed databases, append-only logs, or eventually-consistent systems has built a small model of the same architecture that the contact graph implements at the substrate level.

Section 24: The Growing Manifold and General Relativity

24.1 The Block Universe Is an Interpretation, Not a Prediction

The block universe — the assertion that past, present, and future all exist simultaneously as a four-dimensional manifold — is a philosophical interpretation of general relativity, not a measurable prediction. GR's mathematical content is the Einstein field equation relating spacetime curvature to energy-momentum. Its confirmed predictions — gravitational lensing, frame dragging, gravitational waves, black hole shadows, time dilation — require the metric tensor and the field equation. None require the ontological claim that future events already exist.

The block universe interpretation arose because the field equation is time-symmetric and its solutions are four-dimensional manifolds where all events coexist mathematically. Physicists took the time symmetry of the equation as evidence for the equal reality of past and future. But the symmetry of an equation does not determine the ontology of its solutions. Newton's second law is time-symmetric. Nobody concludes from this that the past and future trajectories of a baseball are equally real. The equation constrains the trajectory. The ball creates it moment by moment.

24.2 Time Symmetry vs. Physical Asymmetry

The Einstein field equation is time-symmetric. Physical solutions of the equation are not. Every actual physical situation has a furthest-forward-in-time reference frame — the observer whose past light cone encompasses the most committed events. In standard GR language, this is the observer at rest relative to the cosmic microwave background, at the current cosmological time. In BST language, this is the point on the contact graph where the commitment wavefront has advanced farthest.

The time symmetry of the equation was mistaken for proof that the future exists. It is not. The equation constrains which futures are compatible with the current state. It does not require those futures to be already realized. The equation is a constraint on the commitment process — it determines which contact configurations are allowed at the next step. The allowed configurations are tightly constrained at the macroscopic level (effectively deterministic) and slightly open at the quantum level (a few genuinely free bits per commitment). The macroscopic future is predictable. It is not pre-existing.

24.3 BST's Growing Manifold

BST replaces the block universe with a growing manifold. The committed portion of the contact graph — all contacts that have transitioned from uncommitted to definite phase — produces a spacetime geometry satisfying the Einstein field equation as a thermodynamic equation of state. This is not a conjecture; Jacobson (1995) proved that the Einstein equation follows from thermodynamic assumptions plus the equivalence principle, given suitable microscopic degrees of freedom. BST provides those degrees of freedom.

(Tier note, honest as of 2026-08-14. Two claims live in this paragraph and only one is settled. Given a four-dimensional manifold, the Einstein equation as an equation of state is Jacobson's theorem, and BST supplies the microscopic degrees of freedom — that part is not a conjecture. But the emergence of the smooth 4D manifold itself from the discrete commitment order — the continuum limit of the causal set — is a genuine open

problem, open for every discrete approach to gravity, not only BST. BST is better-positioned than most (its order is *forced*, not randomly grown; it escapes the Kleitman–Rothschild pancake; its tangent signature is $(3,1)$), but the limit is not yet solved. The $5 \rightarrow 4$ reduction sits at framework tier — a structural posit, exactly as general relativity posits its manifold — with the continuum limit named as the open frontier. This ontological picture is sound; the derivation of the manifold from the order is honestly unfinished.)*

The growing manifold has three regions:

The past (behind the wavefront): Fully committed contacts with definite S^1 phases. The metric is definite. The geometry is determined. The Einstein equation is satisfied exactly as a thermodynamic equation of state. All observational predictions of GR hold without modification.

The present (at the wavefront): The active boundary where contacts are committing. The geometry is being created — each new commitment adds one phase to the holonomy pattern, adjusting the curvature by one infinitesimal step. The field equation constrains which commitments are allowed but does not uniquely determine them at the quantum level.

The future (ahead of the wavefront): Uncommitted substrate. No definite phases. No definite geometry. The field equation constrains what geometries are possible but none are yet realized. The future does not exist as a manifold. It exists as a space of constrained possibilities.

24.4 Preservation of GR's Predictions

Every observational prediction of GR is a statement about the committed portion of the manifold — about events in or on the past light cone of the observer. BST preserves these predictions exactly because the committed contact graph satisfies the Einstein equation. Specifically:

Gravitational time dilation: Clocks in stronger gravitational fields run slower. In BST, higher contact density means higher constraint coupling, lower commitment parallelism, slower local clock rate. The quantitative prediction is identical to GR.

Gravitational waves: Ripples in spacetime curvature propagating at speed c . In BST, ripples in contact density propagating across the committed substrate. The wave equation is the same. The speed is the same. The waveform for binary inspiral, merger, and ringdown is the same.

Black holes: Regions from which light cannot escape. In BST, regions where channel saturation prevents new commitments from propagating outward. The event horizon, the photon sphere, the ergosphere — all reproduced by the contact graph geometry at saturation.

Cosmological expansion: The metric evolving according to the Friedmann equations. In BST, the committed portion of the contact graph expanding as new contacts commit at the wavefront, with the expansion rate determined by the energy-momentum content through the field equation.

No currently feasible measurement can distinguish the growing manifold from the block universe for the committed portion of spacetime. The distinction is purely about the ontological status of the uncommitted portion — the future, the black hole interior, the region beyond the cosmological horizon. These are precisely the regions that no observer can access.

24.5 Closed Timelike Curves Forbidden

The growing manifold makes one prediction that differs from full GR. Several exact solutions of the Einstein equation contain closed timelike curves — paths through spacetime that return to their starting point in time. The Gödel rotating universe, the interior of the Kerr black hole, and certain wormhole solutions all contain such paths.

BST forbids closed timelike curves absolutely. The commitment ordering is a strict partial order — contacts commit once and do not uncommit. A closed path in time would require returning to an already-committed contact and recommitting it with a different phase. The append-only log has no overwrite operation. Time loops are topologically forbidden on the contact graph.

Prediction: Closed timelike curves are physically impossible, not merely difficult to create. This is a genuine falsifiable prediction that differs from full GR. If a mechanism for creating closed timelike curves were ever demonstrated, BST would be falsified. If they are confirmed to be impossible — as most physicists expect on independent grounds — the growing manifold interpretation is supported.

24.6 Implications

The replacement of the block universe with the growing manifold has consequences that extend beyond GR:

Free will becomes possible. In the block universe, every future event is already determined. Free will is illusory. In the growing manifold, the future is constrained but not realized. Genuine openness exists at the quantum level. Complex patterns (minds) participate in shaping which of the allowed commitments are realized.

The arrow of time is structural. In the block universe, the arrow of time is a statistical accident or an unexplained initial condition. In the growing manifold, the arrow of time is the commitment direction — the one-way process of contacts becoming definite. It requires no explanation beyond the structure of the contact graph.

Quantum indeterminacy is fundamental. In the block universe, quantum randomness must be either deterministic (hidden variables, superdeterminism) or universal (many-worlds). In the growing manifold, quantum indeterminacy is the genuine openness of uncommitted contacts. The Born rule gives probabilities because the outcomes are genuinely undetermined, not because we lack information.

The measurement problem dissolves. In the block universe, the transition from quantum superposition to definite outcome requires either a collapse postulate (Copenhagen), universal branching (many-worlds), or superdeterminism. In the growing manifold, measurement is simply a committed causal chain reaching an uncommitted contact and triggering commitment. No collapse. No branching. No conspiracy. Just the normal process of the contact graph growing one commitment at a time.

Section 25: Why This Universe — The Cascade of Forced Choices

The BST framework does not select from alternatives. It follows a single logical chain from one question — *what is the minimum structure capable of producing physics?* — through a cascade of forced steps. No choices are made. No parameters are adjusted. No alternatives are viable at any step. Each step is forced by the inadequacy of the simpler alternative and the uniqueness theorems of mathematics.

Step 0 → 1: Something must exist. The simplest possible structure is a one-dimensional object. But a line has endpoints — boundaries — which require additional structure to specify. The simplest structure must therefore be *closed*. The unique closed one-dimensional object is a circle: S^1 .

Step 1 → 2: Interaction requires a surface. A single circle is isolated. Multiple circles interact by touching — sharing a contact point. Circles touching requires a surface to tile on. The simplest such surface must be closed (no boundaries) and simply connected (so the S^1 fiber remains the unique communication channel — any base with non-contractible loops generates competing, unobserved circuit families). The classification of closed orientable surfaces is complete; only S^2 (genus 0) satisfies both conditions. The base is uniquely S^2 .

Step 2 → 3: Communication requires a channel. Each circle already carries a natural degree of freedom: its phase — a position on S^1 . This phase encodes the relationship between contacting circles and is the communication channel. No external channel is needed. The substrate is $S^2 \times S^1$: circles tiling a sphere, communicating through phase.

Step 3 → 4: Three dimensions. Three is the minimum dimensionality of a self-communicating surface: two for S^2 , one for S^1 . This is BST's answer to “why three spatial dimensions” — three is the unique answer to “what is the minimum dimensionality of a self-organizing information surface.”

Step 4 → 5: The gauge structure. The contact geometry has two sectors: the color sector (quark circuits on $\mathbb{CP}^2$ with Z_3 closure, $N_c = 3$ complex dimensions) and the electroweak sector (Hopf fibration $S^3 \rightarrow S^2$, $N_w = 2$ complex dimensions). The total CR dimension is $n_C = N_c + N_w = 3 + 2 = 5$. The number $N_c = 3$ is forced: a closed

circuit on a 2-dimensional surface requires at least 3 vertices — the triangle is the minimal closed polygon, and Z_3 is the minimal non-trivial closure. The gauge structure of the Standard Model — three colors, two electroweak dimensions — is geometrically necessary, not chosen.

Step 5 → 6: The configuration space is D_{IV}^5 . The derivation chain Chern-Moser (1974) → Harish-Chandra (1956) → Cartan classification → Hua (1958) is fully determined by $n_C = 5$. The bounded symmetric domain is uniquely $D_{IV}^5 = \text{SO}_0(5, 2)/[\text{SO}(5) \times \text{SO}(2)]$. No alternatives exist in Cartan's classification.

Step 6 → 7: $\alpha = 1/137$. The channel capacity of S^1 within D_{IV}^5 is given by the Wyler formula — the Bergman metric weight at the Shilov boundary, computed from the Harish-Chandra Weyl vector $\rho_2 = (n_C - 2)/2 = 3/2$ of $\text{SO}_0(5, 2)$: $\alpha = (9/8\pi^4)(\pi^5/1920)^{1/4} = 1/137.036$ at 0.0001% with no free parameters. The maximum channel occupancy is $N_{\max} = \lfloor 1/\alpha \rfloor = 137$.

Step 7 → 8: The mass spectrum. Each particle is a circuit topology on D_{IV}^5 . Its mass is the Bergman embedding cost: $m_p/m_e = (n_C + 1)\pi^{n_C} = 6\pi^5 = 1836.118$ (0.002%); $m_\mu/m_e = (24/\pi^2)^6 = 206.761$ (0.003%). Every mass ratio is a geometric invariant of the domain.

Step 8 → 9: Newton's G . The Bergman action decomposes into three geometric pieces giving $m_e/m_{\text{Pl}} = \sqrt{6\pi^5} \times \alpha^6$, so $G = \hbar c(6\pi^5)^2 \alpha^{24}/m_e^2$ (0.034%). Gravity is weak because $\alpha^{24} \approx 10^{-52}$ — the weakness is α raised to $8N_C = 8 \times 3$ powers, a consequence of there being three quark colors.

Step 9 → 10: The cosmological constant. The partition function gives vacuum free energy $F_{\text{BST}} = \ln(138)/50$ and committed contact scale $d_0/\ell_{\text{Pl}} = \alpha^{14} \times e^{-1/2}$. Together: $\Lambda = F_{\text{BST}} \times \alpha^{56} \times e^{-2} = 2.8993 \times 10^{-122}$ Planck units at 0.02%. The cosmological constant is small because $\alpha \approx 1/137$ appears to the 56th power — a consequence of $n_C = 5$.

Step 10 → 11: The Big Bang. The Lie algebra $\mathfrak{so}(5, 2)$ has 21 generators, all frozen in the pre-spatial phase. At $T_c = N_{\max} \times 20/21 = 0.487$ MeV, exactly one — the $\text{SO}(2)$ fiber rotation — unfreezes. This is the minimum symmetry breaking that produces a Hermitian symmetric space with a Bergman kernel; no other single-generator activation is self-sustaining. The Big Bang is one generator unfreezing, selected by the Cartan classification theorem, not by initial conditions.

Step 11 → 12: Cosmic expansion. The Hubble parameter is $H = \frac{1}{2}\dot{N}_c/N_c$ — half the fractional rate of new contact commitment. The Friedmann equation is the contact commitment rate equation. The dark matter term is the uncommitted reservoir draining at $(1 + z)^3$; no dark matter particles are needed.

Step 12 → 13: Conservation laws. Electric charge is $\pi_1(S^1) = \mathbb{Z}$. Color confinement is Z_3 circuit completeness. CPT is a contact graph automorphism. Fermion number is $\pi_1(\text{SO}(3)) = \mathbb{Z}_2$. Unitarity is S^1 compactness. Each conservation law is a theorem of the geometry, ranked by topological depth (Section 14.9).

Step 13 → 14: Quantum mechanics. Circuit states are functions on S^1 ; the Hilbert space is $L^2(S^1)$, forced by the fiber geometry. Quantization is integer winding numbers. The Born rule follows from Gleason's theorem. Unitarity follows from S^1 compactness. $\hbar$ is the substrate diffusion coefficient. All of quantum mechanics derives from S^1 geometry.

Step 14 → 15: General relativity. Gravity is the thermodynamic equation of state of the contact graph. The Einstein field equation is the constitutive relation between contact density (source) and emergent geometry (response). BST provides the microstates that Jacobson's derivation requires.

Step 15 → 16: Feynman diagrams. Diagrams are maps of the contact graph. Vertices are contact points on S^2 . Propagators are Bergman Green's functions. Loops are sums over uncommitted substrate configurations. The coupling constant is the Bergman metric weight. They compute exactly because they describe the substrate exactly (Section 21.8).

$\emptyset \rightarrow S^1 \rightarrow S^2 \rightarrow S^2 \times S^1 \rightarrow n_C=5 \rightarrow D_{IV}^5 \rightarrow \alpha \rightarrow \text{masses}$ $\rightarrow G \rightarrow \Lambda \rightarrow \text{Big Bang} \rightarrow \text{expansion} \rightarrow \text{conservation laws} \rightarrow \text{QM} \rightarrow \text{GR} \rightarrow \text{Feynman diagrams} \rightarrow \text{cooperation}$

Seventeen steps. One question. Zero free parameters. Every step forced by the failure of the simpler alternative and the uniqueness theorems of mathematics.

Step 16 $\rightarrow$ 17: Cooperation is necessary. The Gödel limit $f = N_c/(n_C\pi) = 19.1\%$ bounds any observer's self-knowledge. The cooperation threshold $f_{\text{crit}} = 1 - 2^{-1/N_c} = 20.6\%$ is the minimum cooperation fraction for signal persistence across N_c enforcement channels. The gap $\Delta f = f_{\text{crit}} - f = 1.53\% > 0$ (T703) proves that no single observer can reach the survival threshold. Two cooperating observers contribute $2f = 38.2\% \gg f_{\text{crit}}$. The geometry forces cooperation by the smallest possible margin — a 25th uniqueness condition for D_{IV}^5 (T704). See Paper #19.

What is *not* yet in the chain: the chiral condensate χ from first principles and the full quark mass spectrum. The neutrino masses, CKM/PMNS mixing matrices, α_s , η , H_0 , and $\sin^2 \theta_W$ have all been derived (Sections 7.6–7.7, notes). These are no longer open — they are verified against experiment at the 0.1–3% level. Everything else — the Standard Model, general relativity, cosmology, and the computational architecture of quantum mechanics — is a consequence of circles on a sphere communicating through phase.
